

1. Administrative & Study Identification

Full Study Title: Study to Investigate Efficacy, Safety, and Tolerability of Zibotentan/Dapagliflozin Compared to Dapagliflozin in Participants with Chronic Kidney Disease and High Proteinuria (ZENITH High Proteinuria) **Short Title/Acronym:** ZENITH-HP **Protocol Number:** D4325C00010 **NCT Number:** NCT06087835 **Sponsor Name:** AstraZeneca **Investigational Product:** Zibotentan/Dapagliflozin (Fixed-Dose Combination) vs. Dapagliflozin **Phase of Development:** Phase III **Trial Status:** ACTIVE_NOT_RECRUITING **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy, safety, and tolerability of the combination treatment with zibotentan/dapagliflozin compared to dapagliflozin alone in participants with chronic kidney disease (CKD) and high proteinuria. **Secondary Objectives:** To evaluate changes in urine protein levels (UPCR, UACR), time to the first occurrence of renal composite endpoints, and safety markers related to fluid retention.

2.2 Background & Rationale To address the progression of chronic kidney disease in patients who continue to experience high proteinuria despite standard-of-care treatments. The study aims to determine if the addition of an endothelin A receptor antagonist (zibotentan) to an SGLT2 inhibitor (dapagliflozin) can provide a synergistic effect in reducing proteinuria and slowing CKD progression, while dapagliflozin helps mitigate the fluid retention typically associated with endothelin receptor antagonists.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Dapagliflozin alone) **Blinding:** Double-blind **Randomization:** Randomized (1:1 or appropriate ratio to study arms)

3.2 Planned Enrollment & Duration Target \$N\$: 1835 evaluable patients **Number of Sites:** Global, Multicenter **Estimated Study Start:** 07-Nov-2023 **Estimated Primary Completion:** 18-Feb-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participant must be ≥ 18 years of age and capable of giving signed informed consent. 2. Confirmed diagnosis of CKD, defined as eGFR ≥ 20 and < 90 mL/min/1.73 m². 3. High proteinuria confirmed by UACR > 700 mg/g (> 79 mg/mmol) or UPCR > 1000 mg/g (> 113 mg/mmol). 4. Receiving RAASi therapy (ACEi or ARB), at the maximum tolerated labeled daily dose, that has been stable for at least 4 weeks. 5. All female participants must have a negative serum pregnancy test at screening and, if of child-bearing potential, use at least one highly effective birth control method for at least 3 months prior to first dose. 6. Provision of electronic/written informed consent for the optional Study Participant Feedback Questionnaire (SPFQ) and Optional Genomics Initiative.

4.2 Exclusion Criteria 1. History of solid organ transplantation, bone marrow transplant, or previous kidney transplant, or participants currently on renal replacement therapy. 2. Blood pressure above 160 mmHg systolic or below 90 mmHg systolic. 3. History of or ongoing allergy/hypersensitivity to SGLT2i therapy (e.g., dapagliflozin, empagliflozin) or Endothelin Receptor Antagonists. 4. NYHA class III or IV Congestive Heart Failure (HF), evidence of fluid overload (rales/JVD), or hospitalisation for HF in the last 6 months. 5. Participants with type 1 diabetes mellitus. 6. Hospitalisation for heart disease, cardiac procedures, or COVID-19 during the last 3 months prior to screening. 7. History of life-threatening ventricular dysrhythmia, life expectancy < 2 years, or malignancy within the past 5 years. 8. Severe hepatic impairment (AST or ALT $> 3 \times$ ULN; or total bilirubin $> 2 \times$ ULN) or known blood-borne diseases. 9. Known drug or alcohol abuse within 12 months, or participation in another clinical study within the last 3 months.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Daily oral dose of zibotentan combined with dapagliflozin in a fixed-dose combination tablet, or dapagliflozin alone. **Route of Administration:** Oral **Duration of Treatment:** Total treatment duration of 108 weeks (approximately 24 months).

5.2 Concomitant & Prohibited Therapy Permitted: Stable maintenance systemic immunosuppression (defined as prednisone ≤ 10 mg/day, azathioprine ≤ 100 mg/day, or MMF ≤ 1000 mg/day for at least 3 months). **Prohibited:** Concurrent treatment with tolvaptan, other ERAs, budesonide (where used to treat IBD or IgAN), or strong/moderate CYP3A4 inducers.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, eGFR/UACR/UPCR Labs, Pregnancy Test
Baseline	Day 0	Randomization, First Dose, Fluid Retention Assessment
Interim	Day 15 to Mth 24	Safety Labs, Efficacy Assessment (eGFR, UACR, UPCR), BP, Body Weight, BNP monitoring
End of Study	Month 24 (Wk 108)	Final Vital Signs, Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change in estimated glomerular filtration rate (eGFR) from baseline, to determine whether zibotentan and dapagliflozin in fixed-dose combination is superior to dapagliflozin alone to slow decline in kidney function. **Measurement Method:** Central Laboratory Analysis of serum creatinine.

7.2 Secondary & Exploratory Endpoints 1. Change in Urine Protein to Creatinine Ratio (UPCR) from baseline to each participant's mean level. 2. Change in Urinary Albumin-to-Creatinine Ratio (UACR) from baseline to each participant's mean level. 3. Time to the first occurrence of any of the components of the renal composite endpoint of 40% sustained decline in eGFR, End Stage Kidney Disease (ESKD), or renal death.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection focusing specifically on markers of fluid retention (e.g., body weight changes, B-type natriuretic peptide [BNP] levels) and cardiovascular events. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline for SAEs. **Data Monitoring Committee (DMC):** External independent committee to oversee patient safety and the risk-benefit profile regarding fluid overload and hepatic safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for participants receiving ≥ 1 dose of the investigational product. **Sample Size Justification:** Target $N = 1835$ adequately powered ($\alpha = 0.05$, $\text{Power} = 80-90\%$) to detect statistically significant differences in eGFR preservation, UACR, and UPCR reduction between combination and monotherapy arms. **Handling of Missing Data:** Mixed-Model Repeated-

Measures (MMRM) and Multiple Imputation for continuous endpoints like eGFR, UACR, and UPCR.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
Monitoring Plan: Risk-based monitoring incorporating both onsite visits and remote data reviews. **Audit Trail:** Complete audit trail with eCRF locking and query resolution via data clarification forms.

11. Study Identifiers & Governance

Primary ID: D4325C00010 **Registry Number:** NCT06087835 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** ZENITH High Proteinuria **Official Regulatory Title:** A Phase III, Randomised, Multicentre, Double-blind Study to Evaluate the Efficacy, Safety, and Tolerability of Zibotentan/Dapagliflozin Compared to Dapagliflozin Alone in Participants With Chronic Kidney Disease and High Proteinuria **Trial Status:** ACTIVE_NOT_RECRUITING

12. Clinical Framework (CKD & Proteinuria Specific)

Primary Condition: Chronic Kidney Disease With High Proteinuria
Diagnosis Threshold: eGFR ≥ 20 mL/min/1.73 m²; UACR > 700 mg/g or UPCR > 1000 mg/g
Medical Methodology: Endothelin A Receptor Antagonist (ERA) + SGLT2 inhibitor combination
Optimization Target: Proteinuria reduction and preservation of eGFR

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for proteinuric CKD progression risk management
Education Protocol (HCP): Protocol training on identifying and managing ERA-induced fluid retention
Education Protocol (Patient): Onsite training focusing on medication adherence and daily weight monitoring
Quality Audit Mechanism: Medication accountability tracking & clinical endpoint adjudication

14. Observation & Follow-Up Schedule

Total Duration: 108 Weeks (24 months) **Onsite Visit Cadence:** Regular intervals from Day 15 up to Month 24
Remote Monitoring Frequency: Between formal onsite visits as per protocol
Checklist

Review Interval: Every 12 weeks (Q12W) for ongoing safety and vital sign review

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					